

11

STATISTICS AND PROBABILITY

PROBABILITY

Many people suffer from aviophobia, the fear of flying. However, we are more likely to be involved in a car accident than a plane crash. There is a 1-in-5500 chance of being killed on the road compared to 1-in-3 million in the air. This is amazing when you consider that each day worldwide there are 102 000 planes in the sky carrying over 24 million passengers (that's the population of Australia in the sky!). Why do you think there are so few plane accidents compared to road accidents? And why are people more afraid of flying than travelling in a car?



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Chapter outline

	Working mathematically				
11.01 Sample spaces	U	F		R	C
11.02 Probability	U	F		R	C
11.03 The range of probability	U	F		R	C
11.04 Observed and expected frequencies	U	F		R	C
11.05 Complementary events	U	F		R	C

Wordbank

certain Sure to happen; the opposite of impossible

complementary event The 'opposite' event; for example, the complementary event to selecting an Ace from a deck of cards is *not* selecting an Ace

equally likely To have exactly the same chance of occurring; for example, when tossing a coin, heads or tails are equally likely

event A result involving one or more outcomes; for example, the event of rolling an even number on a die contains the 3 outcomes {2, 4, 6}

expected frequency The expected number of times an event will occur over repeated trials

outcome A single result from a situation or experiment; for example, one outcome when rolling a die is to roll a 6

random Describes a situation in which every possible outcome has an equal chance

sample space The set of all possible outcomes of a situation or experiment

Quiz
Wordbank 11

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (4):

11.01 Sample spaces

11.04 Experimental probability • Observed and expected frequencies

11.05 Complementary events

***Twig* video (1):**

11.04 Probability: irrational fears

***PhET* interactive (1):**

11.04 Plinko probability

Quizzes (5):

- Wordbank 11
- SkillCheck 11
- Mental skills 11
- Language of maths 11
- Test yourself 11

Worksheets (14):

SkillCheck Chance cards • Describing probabilities

11.02 Probability problems • Games of chance

11.04 A page of spinners • Coins probability • Dice probability • Coin-toss experiment • Experimental probabilities • Relative frequencies • Spinning chance • A page of spinners

11.05 Theoretical probabilities

Language of maths Probability review

Mind map Probability

Puzzles (8):

11.02 Spinner game • Theoretical probabilities • Basic probability

11.03 Greedy Pig game

11.04 Experimental probability • Duelling dice • Spinner game

11.05 Complementary events

Language of maths Probability crossword

Spreadsheets (2):

11.04 Die rolling simulator • Coin tossing simulator

Presentation (1):

11.04 Theoretical probability

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- list the outcomes of the sample space of a chance experiment
- use the sample space to calculate the probability of an event
- recognise that the range of probability is from 0 to 1
- compare the observed frequency of an event with its expected frequency
- solve problems involving complementary events

Quiz
SkillCheck 11

Worksheets
Chance cards

Describing probabilities

SkillCheck ANSWERS ON P. 602

1 Rate each event as being impossible, unlikely, even chance, likely or certain.

a Choosing a pink ball from a bag of blue balls

b It will rain next month.

c A coin shows tails when you toss it.

d You obtain a driver's licence tomorrow.

e You will have your driver's licence in 20 years' time.

f Rolling double 6s on a pair of dice

g You are over 10 years old.

h You draw a red card from a deck of playing cards.

i You will use a computer today.

j It will snow in your town tomorrow

2 Simplify each fraction.

a $\frac{2}{10}$ **b** $\frac{8}{18}$ **c** $\frac{27}{30}$ **d** $\frac{15}{80}$

3 Convert each number to a decimal.

a $\frac{7}{8}$ **b** $\frac{3}{5}$ **c** 26% **d** 70%

4 Convert each number to a percentage.

a 0.6 **b** 0.31 **c** $\frac{3}{4}$ **d** $\frac{1}{20}$

5 Is the chance of each event more than or less than $\frac{1}{2}$?

a You having another brother or sister one day

b You going overseas this year

c You being at school next Monday

d You being a parent in 20 years' time

e You getting a good report for English this year

f Your home phone ringing today

6 **a** Draw an interval 15 cm long and use it to make a probability scale for the chances from impossible to certain.



b On your scale, mark the position of each event described below.

A It will be hot tomorrow.

B The Sun will rise tomorrow.

C There will be floods in your town this year.

D There will be no car accidents in Sydney tomorrow.

E There are more than 5 people at your home at the same time today.

F Someone at school has a birthday today.

G It will rain tomorrow.

H The Tigers will win their next game.

I The next person to visit your classroom is male.

J Your favourite song comes on your radio station in the next hour.

7 Which term best describes the chance that the next baby born in Australia is a girl? Select the correct answer **A**, **B**, **C** or **D**.

A certain **B** definite **C** even chance **D** probable

8 Convert each number to a simplified fraction.

a 0.35 **b** 0.2 **c** 48% **d** 6%

9 Evaluate each expression.

a $1-\frac{1}{6}$ **b** $1-\frac{3}{4}$ **c** $1-\frac{4}{10}$

11.01 Sample spaces

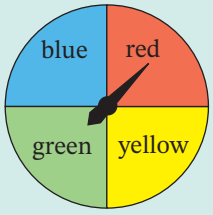
Probability is the branch of mathematics that studies the chances of different events occurring. In any situation, the set of all possible outcomes is called the **sample space**. For example, if a coin is tossed, the sample space is {heads, tails}. If each outcome has an equal chance, then we say that each outcome is **equally likely**.

Example 1

a Write the sample space for this spinner.

b How many outcomes are possible?

c Is each outcome equally likely?



Solution

a The sample space is {red, yellow, green, blue}.

b There are 4 possible outcomes.

c Each coloured region is equal in size ($\frac{1}{4}$ of the circle), so each outcome is equally likely.

The language of probability

This table lists some important terms associated with probability, and how they apply to Example **1**.

Probability term	In Example 1
An **experiment** (or **chance experiment**) is a situation involving chance that leads to results called outcomes.	Spinning a spinner
A **trial** is one go or run of the experiment.	One spin of the spinner
An **outcome** is the result of an experiment.	The arrow landing on green
The **sample space** is the set of all possible outcomes.	{red, yellow, green, blue}
An **event** is one or more outcomes of an experiment.	The arrow landing on a 'traffic light' colour: ed, yellow or green
In a **random** experiment, every possible outcome has the same chance of occurring.	All spins on this spinner are random because every colour has the same chance

Example 2

a How many outcomes are in the sample space when a die is rolled?

b Is each outcome equally likely?

'Die' is the singular word for 'dice': one 'die', 2 or more 'dice'



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Solution

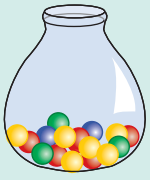
a The sample space is {1, 2, 3, 4, 5, 6}, so there are 6 possible outcomes.

b Each outcome is equally likely, because each number has the same chance of coming up.

Example 3

A jar contains 5 red, 3 green, 6 yellow and 2 blue lollies.
Taylor selects one lolly from the jar at random.

'At random' means that each lolly has an equal chance of being chosen



a How many lollies are in the jar?

b How many outcomes are in the sample space for the colour of the lolly?

c Is each colour equally likely?

Solution

a Number of lollies = 5 + 3 + 6 + 2 = 16

b 4 outcomes {red, green, yellow, blue}

c Each colour is not equally likely because there are different amounts of each colour. Some colours occur more frequently and have higher chances of being chosen.

Video Sample spaces

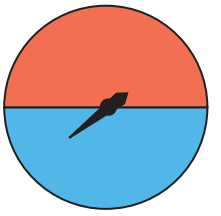
EXERCISE 11.01 ANSWERS ON P. 602

Sample spaces U F R C

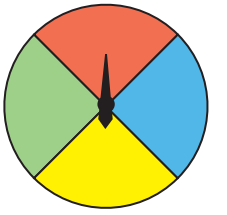
EXAMPLE 1

1 For each spinner, write down the sample space and count the number of possible outcomes. R C

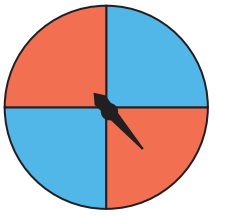
a



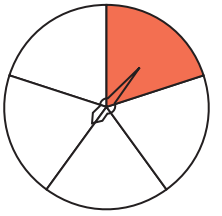
b



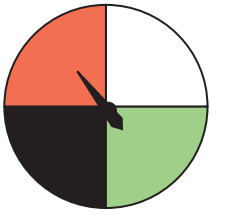
c



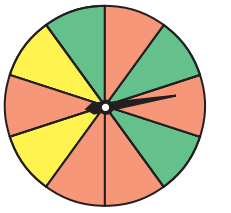
d



e



f



EXAMPLE 2

2 For each experiment, count the number of possible outcomes and state whether each outcome is equally likely.

a Tossing a coin

b The result of a soccer game when Australia plays South Korea

c The first letter of a person's name

d The sex of a baby

e The last digit of a phone number

f The result of a driving test

3 List the outcomes in each event.

a Rolling an odd number on a die

b Selecting a vowel from the letters of the alphabet

c Having a house number greater than 4 but less than 10

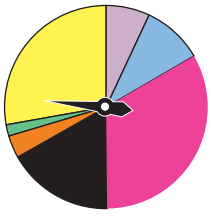
d Having a birthday in a month beginning with M

e Living in a state capital city

f Being in a primary school grade

EXAMPLE 3

4 This spinner is spun. C



a How many outcomes are possible?

b Is each outcome equally likely? Explain your answer.

c Which outcome is the most likely to occur?

d Which outcome is the least likely?

5 A money box contains 4 $2 coins, 3 $1 coins, 2 50c coins, 6 20c coins and 5 10c coins. It is shaken and one coin falls out at random.

a How many coins are in the money box?

b List the outcomes in the sample space for the type of coin.

c Is each type of coin equally likely?

d Which type of coin is most likely to fall out?

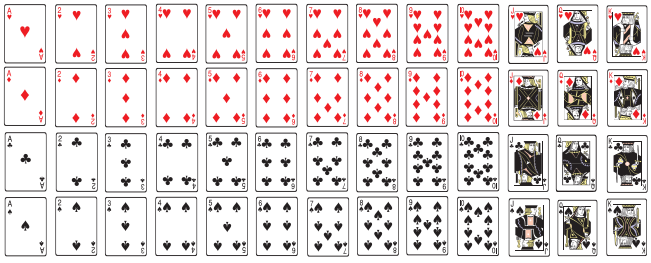
6 The 52 cards in a standard deck of playing cards are shown below, divided evenly into 4 suits: hearts, diamonds, clubs, spades.

Hearts:

Diamonds:

Clubs:

Spades:



The cards are shuffled and one is taken out at random. R

a How many outcomes are in the sample space?

b Is each outcome equally likely?

c How many cards are in each suit?

d How many red cards are in the deck?

e How many 7s?

f How many black Kings?

g How many 8 of diamonds?

h How many cards with an even number?

i How many picture cards (J, K, Q)?

j How many red Aces?

7 A 3-digit number is to be formed from the digits 1, 2, 5. Which of the following answers shows the sample space? Select the correct answer **A**, **B**, **C** or **D**. C

A {1, 2, 5}

B {125}

C {125, 152, 215}

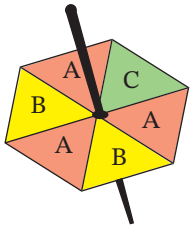
D {125, 152, 215, 251, 512, 521}

8 This spinner is spun. C

a Write down the sample space of possible letters.

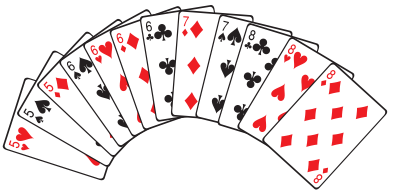
b Is each letter equally likely?

c Which letter is most likely?



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9 Daniel is holding these 12 playing cards. Elvira picks a card without looking. **c**



a How many 5s are held in the 12 cards?

b List the sample space of possible numbers.

c Which number is Elvira most likely to pick?

d List the sample space of possible colours.

e Which colour is Elvira more likely to pick?

f List the sample space of possible suits.

g Which suit is Elvira least likely to pick?

11.02 Probability

We can calculate the chance or probability of an event occurring as a fraction, percentage or a decimal.

Worksheets
Probability problems
Games of chance

Puzzles
Spinner game
Theoretical probabilities
Basic probability

Probability of an event

$P(E)$ means 'the probability of an event, E (occurring)'. If all possible outcomes are **equally likely**, then:

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

$$\text{or } P(E) = \frac{\text{number of outcomes matching } E}{\text{number of outcomes in the sample space}}$$

A **favourable outcome** is one of the outcomes in the event that you want, whose probability you are calculating.

Example 4

A die is rolled. Find the probability of rolling:

a a 5

b a 7

c an odd number

d a number less than 3

Solution

The sample space is {1, 2, 3, 4, 5, 6}. There are 6 possible outcomes and each outcome is equally likely.

a $P(5) = \frac{1}{6}$ — One chance in 6

b $P(7) = \frac{0}{6} = 0$ — No chance, because we cannot roll a 7 on a die

c $P(\text{odd}) = \frac{3}{6} = \frac{1}{2}$ — 3 odd numbers out of 6 {1, 3, 5}

d $P(\text{less than } 3) = \frac{2}{6} = \frac{1}{3}$ — 2 numbers out of 6 {1 and 2}

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Example 5

A jar of lollies contains 5 red, 3 green, 6 yellow and 2 blue lollies. Taylor selects one lolly from the jar at random and notes its colour. Calculate each probability below.



a $P(\text{yellow})$ **b** $P(\text{green or blue})$

c $P(\text{traffic light colour})$, as a percentage **d** $P(\text{not red})$, as a decimal

Solution

Total number of lollies in the jar $= 5 + 3 + 6 + 2 = 16$

a $P(\text{yellow}) = \frac{6}{16}$ $\frac{\text{number of yellow lollies}}{\text{total number of lollies}}$

$= \frac{3}{8}$

b $P(\text{green or blue}) = \frac{3+2}{16}$ $\frac{\text{no. of green lollies + no. of blue lollies}}{\text{total number of lollies}}$

$= \frac{5}{16}$

c $P(\text{traffic light colour}) = P(\text{red or green or yellow})$

$= \frac{5+3+6}{16}$

$= \frac{14}{16}$

$= \frac{14}{16} \times 100\%$ converting to a percentage

$= 87.5\%$

d $P(\text{not red}) = P(\text{green or yellow or blue})$

$= \frac{3+6+2}{16}$

$= \frac{11}{16}$

$= 0.6875$ converting to a decimal

EXERCISE 11.02 ANSWERS ON P. 602

Probability UFRC

1 **a** List all the possible outcomes when a coin is tossed.

b How many different outcomes are possible?

c What is the probability of getting a tail?

d Write the probability of getting a tail as a percentage.



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2 A die is rolled. Find the probability of rolling:

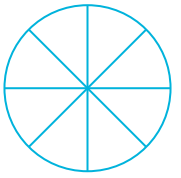
a a 4 **b** an even number **c** a 10

d a number greater than 1 **e** a prime number **f** a number less than 7

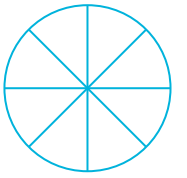
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3 Copy each spinner and shade its regions so that the probability of spinning white is: R

a $\frac{5}{8}$



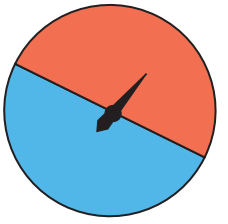
b $\frac{3}{4}$



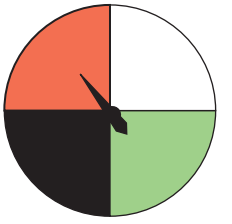
EXAMPLE 5

4 For each spinner, calculate the probability of the pointer landing on red as a percentage.

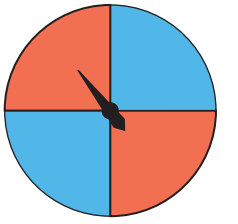
a



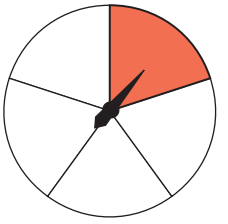
b



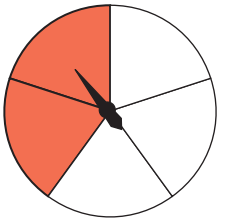
c



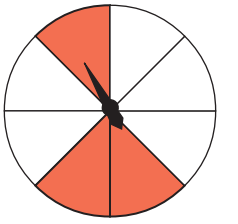
d



e



f



5 A money box contains 4 $2 coins, 3 $1 coins, 2 50c coins, 6 20c coins and 5 10c coins. It is shaken and one coin falls out at random. Calculate each probability:

a P(50c coin)

b P($1 coin), as a decimal

c P(10c or 20c coin), as a percentage

d P(not a 10c coin)

e P(gold coin), as a decimal

f P(a coin under $1), as a percentage

6 A packet of jellybeans has 4 yellow, 3 red, 6 green and 3 black jellybeans remaining. You take out one jellybean at random. What is the probability that it is not a red jellybean? Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{4}{12}$ **B** $\frac{3}{16}$ **C** $\frac{3}{13}$ **D** $\frac{13}{16}$

7 A letter is selected at random from the alphabet. Find the probability that it is:

a M

b R

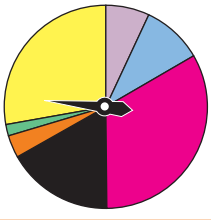
c M or R

d a vowel

e not a vowel

f a letter before K in the alphabet

8 The needle on this spinner is spun.



a Which colour has a probability of $\frac{1}{3}$?

b Which colour has a probability closest to $\frac{1}{4}$?

c Which colour has a probability of $\frac{1}{6}$?

d Which 3 colours have a combined probability of 50%?

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9 A bag contains 6 white, 9 green and 5 blue marbles. Hannah takes one from the bag at random. Which colour has a probability of: R

a 0.25? **b** 45%? **c** $\frac{3}{10}$?

10 A computer generates a random number from 1 to 10. Find:

a P(square number) **b** P(number less than 4), as a decimal

c P(prime number), as a percentage **d** P(factor of 10)

e P(number greater than 6), as a decimal **f** P(5 or 7), as a percentage

11 What is the probability that a baby is born on a weekday rather than a weekend, if each day is equally likely?

12 Design a spinner that would produce these probabilities: R

$P(\text{red}) = \frac{1}{8}$, $P(\text{yellow}) = 24\%$, $P(\text{blue}) = \frac{1}{3}$, $P(\text{black}) = 20\%$.

13 Ryan bought 6 raffle tickets at the school fete. If a total of 800 tickets was sold, what is the probability that Ryan wins first prize? Express your answer:

a as a fraction **b** as a decimal

14 Your maths teacher calls out a name randomly from your class roll. What is the probability that it is:

a your name? **b** a girl's name?

c someone aged 12? **d** someone with blond hair?

15 A deck of 52 playing cards is shuffled and one is taken out at random. Find the probability that it is:

a a red card **b** a club card **c** the King of spades

d a Queen **e** a card with an odd number

f a black picture card **g** a red 7 **h** a 4

You may refer to the diagram of playing cards on page 471.

16 What is the probability that a person randomly chosen has a birthday in a month beginning with the letter J? Select **A**, **B**, **C** or **D**.

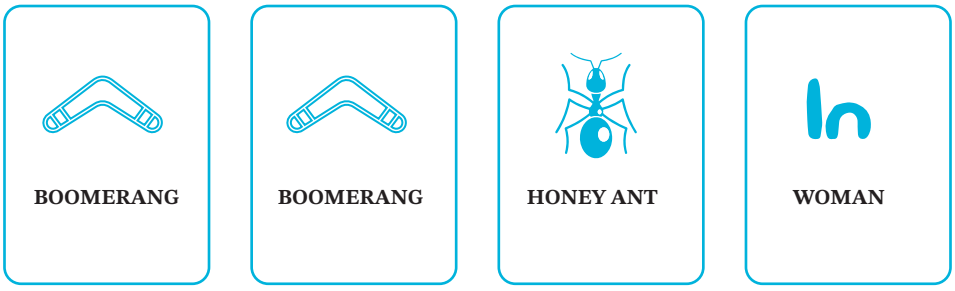
A $\frac{1}{12}$ **B** $\frac{1}{6}$ **C** $\frac{1}{4}$ **D** $\frac{1}{3}$

17 There are 16 teams in a football competition. Paula, therefore, believes that her favourite team, The Bulldogs, has a probability of $\frac{1}{16}$ of winning the competition. Discuss with your friends whether Paula is correct and write your answer, giving reasons. R C

18 Karen had a box of 24 ice blocks. The box had the same number of lemon, orange and strawberry blocks. If she chose one ice block at random, what was the chance that it was either a strawberry or an orange? Select **A**, **B**, **C** or **D**. R

A $\frac{1}{12}$ **B** $\frac{1}{24}$ **C** $\frac{1}{3}$ **D** $\frac{2}{3}$

19 Luca has the following cards with First Nations symbols on them. He chooses 2 cards at random.



a List all possible outcomes.

b What is the probability of choosing

i 2 boomerang cards?

ii one boomerang card and a honey ant card?

iii no boomerang cards?

20 A student council is made up of a number of boys and girls. One member of council is selected at random to attend the regional conference. How many boys and girls could there be in the council if the probability of selecting a boy is $\frac{5}{8}$? R

21 An esky contains 8 cans of lemonade, 5 cans of orange drink and 2 cans of lime drink. How many cans of cola must be added so that the probability of randomly selecting a can of: R

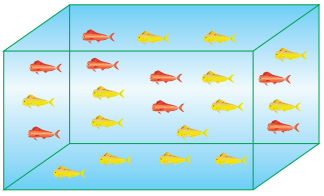
a lemonade is 50%?

b orange drink is 0.25?

c lime drink is $\frac{1}{12}$?

d cola is $\frac{2}{5}$?

22 A fish tank has 8 red and 14 yellow fish.
One fish is randomly selected from the tank. R



a Write the probability that the fish is yellow as a decimal, correct to 2 decimal places.

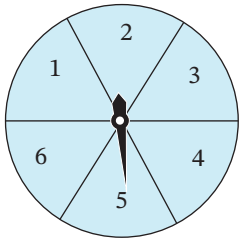
b Write the probability that the fish is red as a percentage.

c How many red fish must be removed from the tank to make the probability of selecting a red fish 0.3?

23 **a** What is the lowest probability value? Name an event that could have this value.

b What is the highest probability value? Name an event that could have this value. R C

24 This spinner is spun twice and the 2 numbers that come up are added. Which of these totals has the lowest chance of occurring? Select **A**, **B**, **C** or **D**. R



A 9 **B** 8 **C** 7 **D** 6

25 A box has 3 pens coloured red and blue. Owen randomly chooses a pen and it is blue. There are now 2 remaining pens in the box. What is the probability that the second pen Owen chooses will be red? (Be careful: the answer is not $\frac{1}{2}$. List all the different possible outcomes.) R C

Foundation Standard Complex

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Did you know?

Car insurance

Insurance is a financial scheme that offers money in times of loss or damage. People can insure their cars against damage or theft. You pay an amount each month or year, and if the loss or damage happens, the insurance company pays you to repair any damage or replace your car.



Shutterstock.com/Dr. Norbert Lange

Insurance was invented in ancient Babylon around 3000 BCE, when voyagers and traders insured their ships against damage or loss. Modern mathematicians working for insurance companies use the statistics of past accidents to predict the chances of future losses. They have found some interesting probabilities involving car accidents.

Event	Chance
Car runs off the road	1 chance in 30
Driver is male	18 chances out of 25
Involves driver fatigue	1 chance in 5
Involves speeding	31 chances out of 100
Involves alcohol	19 chances in 100
Male driver aged 25 to 34	17 chances out of 50

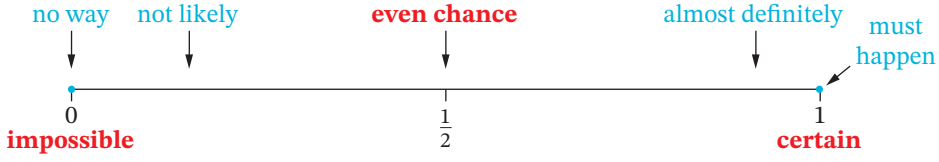
An actuary works in the insurance industry. Find out what an actuary does.

The range of probability 11.03

Because probability is a fraction:

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

its value must range from 0 to 1, or as a percentage, from 0% to 100%.



The range of probability

- The probability of an **impossible** event is 0.
- The probability of a **certain** event is 1.
- The probability of any event ranges from 0 to 1.

A probability close to 1 indicates an event that is very **likely** or probable.

A probability close to zero indicates an event that is very **unlikely** or improbable.

Example 6

5 cards numbered 11, 5, 9, 7 and 3 are shuffled and Jo chooses one at random.
Find the probability that the number chosen is:

a divisible by 3 **b** less than 12

c a factor of 45 **d** even

Solution

a $P(\text{divisible by } 3) = \frac{2}{5}$ 2 numbers {9, 3}

b $P(\text{less than } 12) = \frac{5}{5} = 1$ All 5 numbers are less than 12: a certain event.

c $P(\text{a factor of } 45) = \frac{3}{5}$ 3 numbers {5, 9, 3}

d $P(\text{even}) = \frac{0}{5} = 0$ None are even: an impossible event

EXERCISE 11.03 ANSWERS ON P. 603

Range of probabilities UFRC

EXAMPLE 6

1 If a die is rolled, find the probability of rolling:

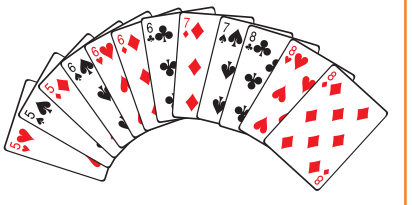
a a 5 **b** an 8 **c** a factor of 12

d a prime number **e** a number less than 7 **f** a multiple of 3

2 One letter is selected at random from the word EVACUATION. Find the percentage probability that it is:

a V **b** A **c** X **d** a vowel

3 Simon is holding these 12 playing cards.
Joshua picks a card without looking.



Find the probability that it:

a is red

b shows an even number

c is a picture card

d has the number 7

e is a hearts card

f not an Ace

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4 A basketball team captain is to be chosen randomly from 4 players, Carl, Lee, Su and Daniela. What is the probability that the captain chosen is:

a Daniela? **b** Su? **c** Manjeet?

5 Match each probability value to its description. C

a $\frac{1}{2}$ **b** 0 **c** 90% **d** 1

e $\frac{3}{4}$ **f** 0.1 **g** 0.6 **h** 2%

A cannot happen **B** better than average chance

C even chance **D** good chance

E very likely **F** almost impossible

G slim chance **H** must happen

6 A coin falls out of a piggy bank containing 4 5c, 5 10c, 6 20c and 3 $1 coins. Find the probability that it is:

a a 5c coin **b** a $1 coin **c** a 50c coin

d a 10c or 20c coin **e** not a $1 coin **f** not a 20c coin

7 A traffic light shows red for 59 seconds, green for 99 seconds and yellow for 2 seconds. Find, as a decimal, the probability that it shows:

a green **b** yellow

c red or green **d** not red, yellow or green

8 A computer generates a random number from 1 to 20. Find, as a percentage, the probability that the number is:

a a multiple of 4 **b** a factor of 20 **c** a number less than 30

d 25 **e** a prime number **f** divisible by 5

9 A die has 2 blue faces, 1 red face and the other faces are green. It is rolled. Match each event to its correct probability:

a red **b** blue

c yellow **d** a colour that is not red

e blue, red or green **f** a traffic light colour

A $\frac{2}{3}$ **B** 50% **C** $\frac{5}{6}$ **D** $\frac{1}{3}$

E 0 **F** $\frac{1}{6}$ **G** 100%

10 A letter is chosen at random from the words 'NEW CENTURY MATHS'. Find each probability.

a $P(\text{T})$ **b** $P(\text{M})$ **c** $P(\text{N or E})$

d $P(\text{a consonant})$ **e** $P(\text{a letter also found in PROBABILITY})$

f $P(\text{K})$

11 Write an event that could have a probability of: R C

a $\frac{1}{2}$ **b** 0 **c** 90% **d** 1

e $\frac{3}{4}$ **f** 0.1 **g** 0.6 **h** 2%

Did you know?

Buckley's chance

When an Australian person says you have 'Buckley's chance' of something happening, it means that you have little chance or no chance at all. This old slang phrase is based on the name of a Melbourne department store in the 1850s called 'Buckley and Nunn'. The original expression was 'You have 2 chances: Buckley's and none'.

English is often a funny and illogical language. For example, 'slim chance' and 'fat chance' sound like opposites, but they both mean the same thing.

Find another slang expression that describes a chance.

Puzzle
Greedy Pig game

Investigation

The Greedy Pig game



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This is a game for 4 or more players.
You need a playing die and a score sheet (download and print out the puzzle sheet 'Greedy Pig game').

1. All players stand at the start of a round. A die is rolled twice and the total is each player's starting score.
2. Before the next roll of the die, each player has the option of sitting down and keeping their score, or remaining standing and 'being greedy' with the chance of increasing their score from the next roll of the die.
3. The die is rolled and the number is added to the score of each player standing, but if 2 comes up, all standing players lose their points.
4. A round ends when a 2 is rolled or all players are seated. Each player then records their score for the round.
5. The round is repeated 4 more times and the overall winner is the player with the highest total score for the 5 rounds.
6. Play this game several times. Is there any strategy to winning?

Observed and expected frequencies 11.04

Probability calculated using the formula:

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

is more specifically called **theoretical probability** or **expected probability**.

We can also determine probability based on the results of an experiment or trial that has been repeated many times, such as testing the safety of different cars, or rely on past statistics, such as the number of rainy days in April. This type of probability is called **observed probability** or **experimental probability**, which is based on **relative frequency**, the number of times an event occurred as a fraction of the total **frequency** of outcomes.

Relative frequency

$$P(E) = \frac{\text{number of times the event happened}}{\text{total number of trials}}$$

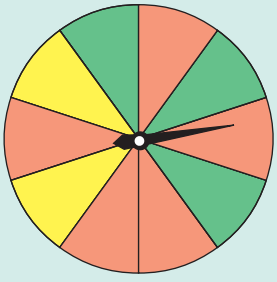
or $P(E) = \frac{\text{frequency of } E}{\text{total frequency}}$

As in statistics, 'frequency' means the number of times something happens.

Example 7

Josie spun this spinner 80 times and found the following results:

Outcome	Red	Green	Yellow
No. of times	44	25	11



a What is the theoretical probability of spinning red?

b For 80 spins, what is the expected number of times of spinning red? How does this compare with the actual number of times?

c What is the observed probability of spinning red?

Solution

a $P(\text{red}) = \frac{5}{10} = \frac{1}{2}$

b Expected number of reds $= \frac{1}{2} \times 80 = 40$.

From the table, the observed number of reds = 44, which is close to the expected number.

c Observed $P(\text{red}) = \frac{44}{80} = \frac{11}{20}$

Worksheets
A page of spinners
Dice probability
Coins probability
Coin toss experiment
Experimental probabilities
Relative frequencies

Puzzles
Experimental probability
Duelling dice

Interactive
Plinko probability

Presentation
Theoretical probability

Videos
Experimental probability
Probability: irrational fears

Observed and expected frequencies

The expected number of times an event will occur over repeated trials is called the **expected frequency**.

Expected frequency = theoretical probability × number of trials

The actual number of times an event occurs over repeated trials in an experiment is called the **observed frequency**.



Video
Observed and expected frequencies

Example 8

Declan rolled a die 60 times and recorded the results in a table.

Outcome	1	2	3	4	5	6
Frequency	10	11	13	7	8	11

a What is the theoretical probability of rolling a 5 or 6 on a die?

b For 60 rolls, what is the expected frequency of rolling a 5 or 6? How does this compare with the observed frequency?

c What is the experimental probability of rolling a 5 or 6?

Solution

a $P(5 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}$

b Expected frequency of 5s or 6s $= \frac{1}{3} \times 60$ Probability × number of trials
$= 20$

From the table, the observed frequency of 5s and 6s $= 8 + 11 = 19$, which is close to the expected frequency, 20.

c Experimental $P(5 \text{ or } 6) = \frac{8 + 11}{60} = \frac{19}{60}$

EXERCISE 11.04 ANSWERS ON P. 603

Observed and expected frequencies U F R C

EXAMPLE 7

1 For the spinner in Example **7**, find: C

a the theoretical probability of spinning yellow

b the expected frequency of spinning yellow and how this compares with the observed frequency

c the observed probability of spinning yellow

2 A pair of dice was rolled 50 times and their sum calculated each time.

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	0	2	4	6	5	5	9	6	8	3	2



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a Find, as a decimal, the observed probability of rolling a sum:

i of 10 **ii** of 7 or 11

iii that is even **iv** less than 6

v greater than 10 **vi** that is a prime number

b Which sum:

i was most likely? **ii** had a probability of $\frac{2}{25}$?

iii was least likely? **iv** had a probability of 10%?

v was second-most likely? **vi** had a probability of $\frac{4}{25}$?

EXAMPLE 8

3 **a** Copy this table.

Outcome	Tally	Frequency
Head		
Tail		
	Total	



Shutterstock.com/James Steidl

b Toss a coin 50 times and record the result for each toss in the table.

c What is the theoretical probability of tossing a tail?

d In 50 tosses, what is the expected frequency of tails? How does this compare with the observed frequency?

e What is the observed probability of tossing a tail?

f What do you think would happen to the observed probability of a tail if the coin was tossed 1000 times rather than 50? R C

4 The volume of soft drink in 50 cans of drink were measured.

Volume (mL)	373	374	375	376	377
Number of cans	2	6	38	3	1

Using these results, if you bought a can of drink, what is the probability that it will contain exactly 375 mL of drink? Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{50}{375}$ **B** $\frac{19}{25}$ **C** $\frac{6}{25}$ **D** $\frac{3}{50}$

5 **a** Copy this table.

Outcome	Tally	Frequency
1		
2		
3		
4		
5		
6		
	Total:	

b Roll a die 50 times and record the result for each roll in the table.

Foundation Standard Complex

c Find the expected probability and expected frequency of rolling:

i 3 **ii** 6

iii an even number **iv** a number below 6.

d Find the observed probability of rolling:

i 3 **ii** 6

iii an even number **iv** a number below 6.

e How do the observed frequencies compare with the expected frequencies?

f What would you expect to happen if the die was rolled 1000 times? R C

6 A die was rolled 80 times, with the results shown below. R C

Outcome	1	2	3	4	5	6
Frequency	11	13	9	13	12	22

a When rolling a normal die, is each outcome equally likely?

b Do you think this die is loaded (unfair)? Give a reason for your answer.

c Write the observed probability of rolling a 1 on this die as a percentage.

d If this die was rolled 100 times, how many times would you expect 3 to come up?

7 Mila tossed a coin many times and got 140 heads and 110 tails. Calculate the observed probability of heads with this coin as a percentage.

8 Luna surveyed all the Year 7 students at her school about their favourite sport.

Which sport did 3 out of every 10 students choose?

A Soccer **B** Swimming

C Netball **D** Hockey

Sport	Number of students
Netball	54
Soccer	75
Hockey	25
Swimming	66

Spreadsheets
Die rolling simulator

Coin tossing simulator

Technology

Rolling a die

Scientists and economists often simulate events on a computer, which saves time and money because large amounts of data can be obtained quickly.
In this activity, a scientific calculator or spreadsheet is used to simulate the rolling of a die.

1 Copy this table.

Outcome	Number of times rolled
1	
2	
3	
4	
5	
6	

Foundation Standard Complex

9780170477963

2 Random numbers from 1 to 6 can be generated on a calculator or spreadsheet using the RanInt, RANDOM or =RAND() functions.

Casio scientific calcula or	Sharp scientific calcula or
Enter the formula RanInt#(1,6) as shown below:	2nd F 7 (RANDOM), select 1 R-DICE, =
ALPHA . 1 Shift) 6) =	

Press = more times to generate more random integers from 1 to 6.

On a spreadsheet, type **=INT(RAND()*6+1)** into a cell and press the Enter key.

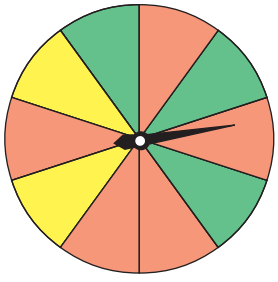
3 Run the simulation 20 times and record the results in your table.

4 Are certain numbers more likely to be rolled than others? (For example, is a 2 more likely to be rolled than a 5?) Do your results reflect this?

5 Compare your results with the simulated results of your class. Are they similar or different? Are they what you and your classmates expected? Discuss.

Investigation

Spinning chance



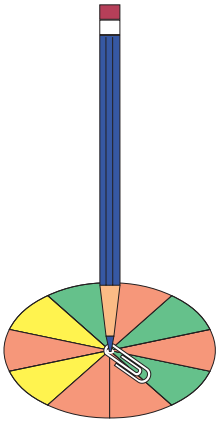
You will need: A pencil, a paper clip and an enlarged copy of the spinner used in Example **7** (print out the worksheet 'Spinning chance').

1 Which colour is the most likely to be spun? Why?

2 Which colour is the least likely?

3 Copy this table.

Outcome	Tally	Frequency
Red		
Yellow		
Green		
	Total:	



4 Work with a partner. Place a pencil on the centre of the spinner and a paper clip around the point. Spin the paper clip around the spinner, and note which colour it lands on.

5 Take turns to spin the spinner 100 times and use the table to record your results.

6 Find the theoretical probability and expected frequency of spinning:

i red **ii** yellow

iii green.

11.04

Worksheets
Spinning chance

A page of spinners

Puzzle
Spinner game

7 How did your results compare with the expected frequency?

8 Find the relative frequency of spinning:

i red **ii** yellow **iii** green.

9 What would happen if 1000 spins were simulated? How many of each colour would be expected?

10 Combine the results of your class on a whiteboard or projector to calculate the observed probability for each colour. For such a large number of trials, are the observed probabilities closer to the theoretical probabilities?

11 The spinner can also be simulated using the random number generator on a calculator, which outputs a random decimal between 0 and 1 with 3 decimal places.

Let 0, 1, 2, 3 and 4 represent red, 5 and 6 represent yellow and 7, 8 and 9 represent green. Now each decimal generated can represent 3 spins of the spinner. For example, 0.561 represents yellow, yellow, red.

12 Draw the blank table from question **8** again and use the 'Random' function on your calculator to simulate 100 spins.

13 Use the table to record your results.

14 How did your results compare with the expected frequencies?

Quiz
Mental skills 11

Mental skills 11: Maths without calculators ANSWERS ON P. 603

Adding 90c and 95c

A quick way to mentally add monetary amounts ending in 90c or 95c is to round up to whole dollars and then subtract 10c or 5c respectively.

1 Study each example.

a $\$8.95 + \$17.95 + \$11.95 = \$9 + \$18 + \$12 - 5c - 5c - 5c$
$= \$39 - 15c$
$= \$38.85$

b $\$14.90 + \$5.95 + \$12.90 = \$15 + \$6 + \$13 - 10c - 5c - 10c$
$= \$34 - 25c$
$= \$33.75$

c $\$18.90 + \$24.90 + \$8.80 = \$19 + \$25 + \$9 - 10c - 10c - 20c$
$= \$53 - 40c$
$= \$52.60$

2 Now simplify each expression.

a $\$12.95 + \$7.95 + \$14.95$
b $\$20.90 + \$4.95 + \$13.95$
c $\$9.95 + \$11.90 + \$13.90$
d $\$11.95 + \$15.90 + \$6.95$
e $\$13.95 + \$19.95 + \$8.95$
f $\$34.90 + \$34.90$
g $\$27.95 + \$16.95 + \$12.90$
h $\$11.90 + \$21.95 + \$19.80$
i $\$24.95 + \$34.95$
j $\$33.90 + \$49.90 + \$49.90$

Investigation

Complementary events

In everyday life, we use the word 'complementary' to describe things that go together and 'complete the picture' when they are together. For example, when dressing for an occasion:

- a shirt and a matching tie complement each other
- a dress and a matching pair of shoes complement each other

Remember also that 'complementary' angles add to 90°.

In probability, **complementary events** are events that together make up all the possible outcomes.

Not to be confused with **complimentary**, which means speaking highly about someone and paying them a compliment or to describe an item received for free.

Event	Complementary event
Tossing a tail on a coin	Tossing a head on a coin
Rolling a 6 on a die	Rolling any of the other numbers, from 1 to 5, on a die
Raining	Not raining
Being born on a Monday	Being born on a day other than Monday

1 Suppose that there is an equal chance of being born on any day of the week: Monday to Sunday.

a What is $P(\text{Tues})$, the probability of being born on a Tuesday?

b What is $P(\text{not Tues})$, the probability of being born on a day other than Tuesday?

c What do you notice about $P(\text{Tues}) + P(\text{not Tues})$?

2 A fruit bowl contains 7 apples, 4 oranges and 9 bananas. One piece of fruit is selected at random from the bowl.

a Find $P(\text{orange})$ **b** Find $P(\text{not orange})$

c What do you notice about $P(\text{orange}) + P(\text{not orange})$?

3 A baby is selected at random from the maternity section of a large hospital. There is an equal chance of the baby being a boy or a girl.

a Find $P(\text{boy})$ **b** Find $P(\text{girl})$

c What do you notice about $P(\text{boy}) + P(\text{girl})$?

4 Copy and complete the following sentence:

The probability of an event _____ the probability of its complementary event must always equal _____.

11.05 Complementary events



Worksheet
Theoretical probabilities

Puzzle
Complementary events



Video
Complementary events

In any situation, the probabilities of all possible outcomes must add to 1.

Complementary events are 2 events that together make up all the possible outcomes, such as a head and a tail when tossing a coin. The **complement** of an event E, are all those outcomes that are *not* E, or that are the 'opposite' of E.

Because an event and its complement covers all possible outcomes, the sum of their probabilities must equal 1.

Complementary events

$P(E) + P(\text{not } E) = 1$

or $P(\text{not } E) = 1 - P(E)$

or $P(\text{complementary event}) = 1 - P(\text{event})$

or $P(\text{event not occurring}) = 1 - P(\text{event occurring})$

If the probability is written in percentage form, then $P(\text{not } E) = 100\% - P(E)$.

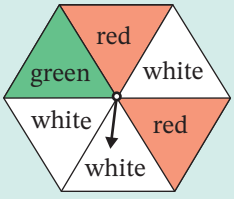
Example 9

On this spinner, what is the probability of spinning:

a red?

b a colour that is not red?

c a colour that is not green?



Solution

a $P(\text{red}) = \frac{2}{6} = \frac{1}{3}$

b $P(\text{not red}) = 1 - P(\text{red})$

$= 1 - \frac{1}{3}$

$= \frac{2}{3}$

The complement of 'red' is 'not red', which is 'white or green'

Note that $P(\text{red}) + P(\text{not red}) = \frac{1}{3} + \frac{2}{3} = 1$, which covers all possible outcomes.

c $P(\text{green}) = \frac{1}{6}$

$P(\text{not green}) = 1 - P(\text{green})$

$= 1 - \frac{1}{6}$

$= \frac{5}{6}$

The complement of 'green' is 'not green', which is 'red or white'

Note that $P(\text{green}) + P(\text{not green}) = \frac{1}{6} + \frac{5}{6} = 1$, which covers all possible outcomes.

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EXERCISE 11.05 ANSWERS ON P. 603

Complementary events U F R C

1 Write the complement for each event. R C

a The sex of a new baby being female

b Rain tomorrow

c Losing a soccer match

d Being left-handed

e Being over 18 years of age

f Choosing a hearts card from a deck of playing cards

g A traffic light showing green

h A train arriving late

EXAMPLE 9

2 A die is rolled. What is the probability that the result is:

a 3? **b** less than 3? **c** prime?

d not a 3? **e** not less than 3? **f** not prime?

3 Olivia buys a ticket in a raffle in which 1000 tickets are sold and there is only one prize. What is the probability of Olivia:

a winning the prize? **b** not winning the prize?

4 A jar contains jellybeans in the following colours and amounts: red 40, blue 25, black 50, white 15. One jellybean is selected at random. What is the probability that the jellybean is:



a white? **b** not white?

c yellow? **d** not yellow?

e not red? **f** not blue or black?

5 Find the probability of the event that is **complementary** to each event described.

a The probability of choosing a Jack from a pack of cards is $\frac{1}{13}$.

b The chance of shooting a basketball hoop is 55%.

c The probability of winning a prize is 0.07.

6 A bag of toy cars contains only red, blue and white cars. If you take out a car at random, the chance of it being red is 0.5, and the chance of it being white is 0.2.

- **a** What is the chance of selecting a red or white car?
- **b** What is the chance of selecting a blue car?
- **c** If the bag holds 30 cars, how many of each colour would you expect to find?
- **d** What is the chance of the car you select being pink?

7 What is the probability that the next person you meet was born in a month:

- **a** beginning with the letter A?
- **b** not beginning with the letter A?

8 4 students, Sue, Liam, Emily and Matt, write their names on cards and place the cards in a bag. A card is chosen, without looking, to select the class captain.

- **a** Find the probability that Emily was not chosen.
- **b** Find the probability that the captain is a boy.
- **c** What is the chance that the captain is not a boy?
- **d** What is the chance that the captain is the teacher?

9 Which of the following is the complementary event to 'winning a race'?
Select the correct answer **A**, **B**, **C** or **D**. R C

A coming last
B coming second or third
C not winning the race
D coming second

10 In a netball match, the Swifts have a 43% chance of winning, while the Vixens have a 49% chance of winning. R C

- **a** What other outcome is possible?
- **b** What is the probability of this outcome?

11 The probability that a person has skin cancer is 0.03. What is the probability that a person doesn't have skin cancer? Select **A**, **B**, **C** or **D**.

A 0.2 **B** 0.7 **C** 0.07 **D** 0.97

12 The letters of the word PROBABILITY are written on separate cards and one is randomly selected. What is the probability that a letter drawn out is:

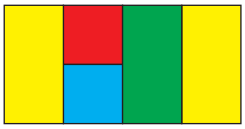
a not P? **b** not a vowel? **c** not I? **d** not A or B?

13 The probability that it will rain this weekend is 85%. What is the probability that it won't rain?

14 A die is rolled. What is the chance that the number rolled is not a factor of 6? Select **A**, **B**, **C** or **D**.

A $\frac{2}{3}$ **B** $\frac{5}{6}$ **C** $\frac{1}{3}$ **D** $\frac{1}{6}$

15 Stef has a rug with this pattern of colours. Ashton drops his ice cream on the rug. What is the probability that the ice cream lands on the section coloured:



a yellow? **b** red?

c blue or green? **d** not green?

16 What is the decimal probability that a mobile phone number selected at random does not end in 0 or 1? R

Power plus ANSWERS ON P. 604

1 2 dice are rolled and the sum of the numbers is calculated.

a Copy and complete this table to show all possible sums.

b How many different sums are possible?

c Why isn't each sum equally likely?

d Which sum is the most likely?

e Which sum is the least likely?

		1st die					
	+	1	2	3	4	5	6
2nd die	1	2	3	4	5	6	7
	2	3	4				
	3	4					
	4						
	5						
	6						

f What is the probability of a sum of 2?

g What is the probability of a sum of 10?

h Which sum has a probability of $\frac{1}{9}$?

2 3 coins are tossed together. One possible outcome is HHT, that is, heads on the first coin, heads on the second coin, tails on the third coin.

a How many other possible outcomes are there? List them.

b Use your answers from part **a** to calculate the probability of:

i 3 heads **ii** 2 heads **iii** one head

iv no heads **v** at least one head **vi** at most one head

3 At Harry's takeaway, you can order a special meal deal for $8 that consists of one main meal, one drink and one dessert.

Harry's Takeaway Menu

Main meals	Drinks	Desserts
Hot dog	*Cola*	*Cake*
Pizza	*Lemonade*	*Ice cream*
Hamburger		



a List all possible meal deal combinations. How many are there?

b If you were to choose a combination at random, what is the probability that it will:

i be pizza, lemonade and cake?

ii not contain a hot dog?

iii contain cola?

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CHAPTER 11 REVIEW

Language of maths

certain	chance	complementary event	die/dice
equally likely	event	expected frequency	experiment
favourable	frequency	impossible	improbable
likely	observed frequency	observed probability	outcome
probability	probable	random	relative frequency
sample space	theoretical probability	trial	unlikely

1 What is the opposite of **impossible** and what is its probability value?

2 What is the difference between **impossible** and **improbable**?

3 What is the name given to the set of all possible outcomes for a situation?

4 What is the **complementary event** to a baby being born on a weekend?

5 What does it mean when a name is drawn out of a box '**at random**'?

6 What word means the number of times something happens?

Quiz
Language of maths 11

Puzzle
Probability crossword

Worksheet
Probability review

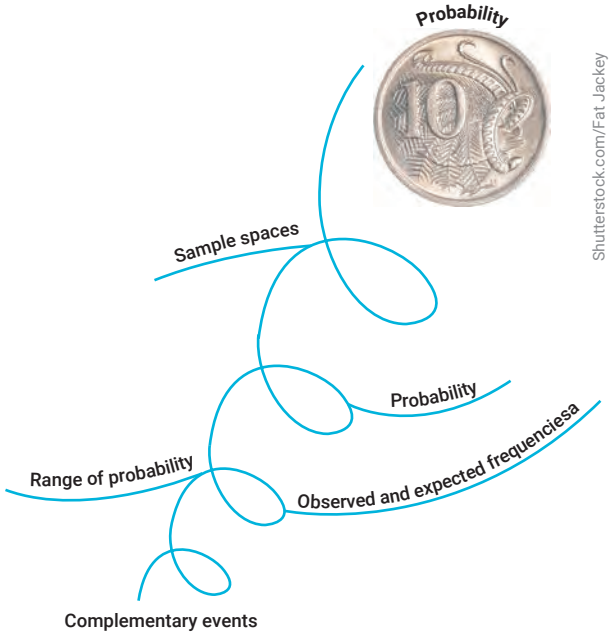
Topic summary

- Write about what you have learnt in this chapter.
- Write down the parts of the chapter that were new to you.
- Copy and complete:

  The things I understand about probability are ...

  The things I am still not confident about in this chapter are ...

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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Worksheet
Mind map: Probability

Quiz
Test yourself 11

TEST YOURSELF 11 ANSWERS ON P. 604

11.01

1 List the outcomes for the sample space in each situation.

a The result of a driving test

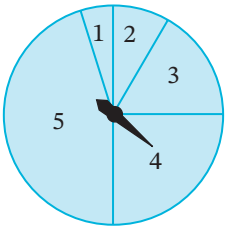
b The classification of a new film (for example, PG)

c The winner of a tennis match when Katrina plays Biljana

d The public transport students use to attend your school

11.01

2 **a** For this spinner, how many outcomes are there in the sample space?



b Why isn't each outcome equally likely?

c Which number is the most likely?

d Which number is the least likely to be spun?

e Which number has a probability of 25%?

11.02

3 On a die, what is the probability of rolling:

a a 1? **b** an odd number? **c** a number less than 5?

11.02

4 A box of chocolates contains 15 dark chocolates, 13 milk chocolates and 12 white chocolates. Markos selects one chocolate at random. What is the probability that he has chosen a white chocolate? Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{1}{28}$ **B** $\frac{1}{12}$ **C** $\frac{3}{10}$ **D** $\frac{7}{10}$

11.02

5 A truck carries 325 boxes of exercise books, 210 boxes of rulers, 360 boxes of paper and 145 boxes of pens. If a box is taken at random from the truck, what is the probability (as a percentage correct to one decimal place) that the box contains:

a rulers? **b** no pens?

c exercise books or paper? **d** erasers?

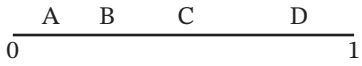


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6 Where on the probability scale is the chance that the next baby born at the local hospital is a girl? Select **A**, **B**, **C** or **D**. 11.03



7 A jar contains 1 red, 6 yellow, 2 white and 5 black jellybeans. If a jellybean is selected at random, find the probability that it is: 11.03

a yellow **b** white or red **c** blue **d** not green

8 Which word best describes an event that has a probability of 0.29? Select **A**, **B**, **C** or **D**. 11.03

A unlikely **B** highly probable

C almost impossible **D** good chance

9 Which event below could be described as a certain event? Select **A**, **B**, **C** or **D**. 11.03

A Choosing a 4 from a hat containing the numbers 1, 2, 3 and 4

B Choosing a red ball from a bag of green balls

C Rolling an odd or even number on a die

D A new baby being a girl

10 A coin was tossed 80 times and landed heads 24 times. 11.04

a What is the theoretical probability of obtaining a head?

b What is the expected frequency of heads from 80 tosses of a coin? How does the observed frequency compare to this?

c What is the observed probability of obtaining a head?

11 Write the complementary event for each event, and its probability. 11.05

a Selecting a queen from a deck of cards.

b Rolling a multiple of 3 on a die.

c Buying the winning ticket out of 550 tickets sold.

12 The names of 5 players; Ashleigh, Hamish, Sarah, Jake and Hayley are written on separate cards and put into a bag. One name is drawn at random from the bag. What is the probability that the name selected: 11.05

a is Hamish? **b** is a girl's name? **c** does not begin with H?

13 There are 8 brown marbles, 4 black marbles and 3 white marbles in a bag. One marble is taken at random from the bag. What is the probability that the marble selected is not brown? 11.05